

1. (40 points) For each of the following inputs $x(t)$, determine the output response $y(t)$ for a system with the following transfer function using the Laplace transform and partial fraction expansions:

$$H(s) = \frac{2s - 1}{s^2 - 2s - 15}$$

a. $x_1(t) = u(t)$

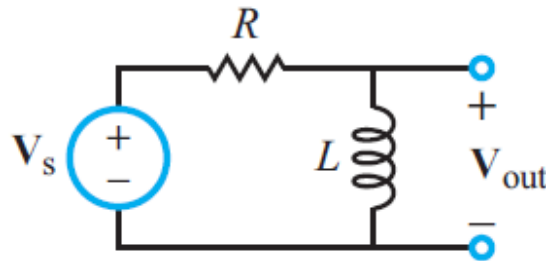
b. $x_2(t) = t u(t)$

c. $x_3(t) = e^{-t}u(t)$

d. $x_4(t) = \cos(t)u(t)$

2. (20 points) Using the Laplace transform, redo the convolution rules 1-5 in Table 2-1 of your textbook.

3. (20 points) Using the Laplace transform, solve the differential equation for the RL circuit to derive the unit-step response given that the initial output voltage is 1V.



4. (20 points) Calculate the inverse Laplace transforms for the following signals.

a. $X_1(s) = 1 + \frac{s+4}{s^2+16}$

b. $X_2(s) = \frac{(s-5)e^{-2s}}{(s+1)(s+7)}$